

Scalability & Future plan

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Team ID: XXXXX

Project Name: Smart Lender – Loan Eligibility Prediction System

Maximum Marks: 1 Mark

Scalability & Future Plan

This document describes how the **Smart Lender – Loan Eligibility Prediction System** can be enhanced to support more users, larger datasets, improved performance, and additional features in the future. As this project was developed individually, the future enhancements represent the planned roadmap for extending the application beyond its current implementation.

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	The application uses a static dataset for prediction.	Predictions may not reflect the latest banking trends or customer profiles.	High
2	No database is used to store user applications or prediction history.	User data cannot be retrieved or managed after submission.	Medium
3	The application does not include user authentication or role-based access.	Access control and personalized user management are not available.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local system.	Deploy the application on a cloud platform (AWS, Azure, or Google Cloud) to support multiple concurrent users.
2	Data Storage	Uses a local dataset and trained model.	Integrate a relational database such as MySQL or PostgreSQL to securely store applicant information and prediction records.
3	Performance	Suitable for small-scale usage.	Optimize model loading, improve backend processing, and use caching techniques to reduce response time.

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
4	Security	Basic input validation is implemented.	Add user authentication, encrypted communication (HTTPS), and role-based access control to enhance security.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Improve the Machine Learning model using larger and more diverse datasets.	Short Term	Increase prediction accuracy and reliability.
Phase 2	Integrate a database to store applicant details and prediction history.	Medium Term	Enable data management and reporting capabilities.
Phase 3	Deploy the application on a cloud platform and implement secure user authentication.	Medium Term	Improve accessibility, scalability, and system security.
Phase 4	Develop a mobile application and integrate real-time credit score and document verification services.	Long Term	Provide a complete digital loan eligibility solution with enhanced user convenience.